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Observability-Aware Virtual Force Sensing and Self-Calibration for Quadrotors

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ABSTRACT Many aerial tasks, such as flying in wind or carrying a payload, require the external force on the airframe, yet quadrotors rarely carry a force sensor. We turn the onboard inertial measurement unit into a *virtual force sensor*: the external specific force is read directly from the accelerometer residual. The obstacle is calibration, since the reading needs the vehicle mass, which is unobservable near hover, where it is indistinguishable from a vertical force; a standard extended Kalman filter then corrupts the mass. We make the sensor *observability-aware and self-calibrating*. An online identifiability measure, the Fisher information on the mass, gates the calibration and freezes the mass when it cannot be seen; and when the flight lacks the excitation to recalibrate, the controller injects a minimal probe that the same measure shapes. In simulation the sensor reads the external force with a per-axis correlation of 0.93 under continuous wind; the gate holds the mass calibration to $3\,\%$ through a windy hover where an ungated filter drifts to $7\,\%$; the probe recovers the mass from a wrong prior that hover alone cannot identify; and feeding the sensed force to a model-predictive controller reduces the tracking error by 34 to $80\,\%$ under wind, most where the wind dominates the dynamics. The sensor degrades gracefully under injected accelerometer and thrust errors and runs in real time at $100\,\mathrm{Hz}$.

INDEX TERMS Aerial robotics, force estimation, inertial sensing, observability, self-calibration, virtual sensors.

I. INTRODUCTION

THE external force acting on a quadrotor, from wind, a payload, or physical contact, governs how it must be flown, yet small aerial platforms rarely carry a force or wrench sensor: the payload and power budget favor reusing the sensors already on board. The accelerometer offers a route. It measures the non-gravitational specific force directly, so once gravity and the commanded thrust are removed, what remains is the external specific force. This turns the inertial measurement unit (IMU) into a *virtual force sensor*, with no added hardware, and the reading is instantaneous: no double-differentiation of position, no estimator lag.

The virtual sensor has one calibration parameter, the vehicle mass, which converts the measured acceleration into the thrust contribution that must be subtracted. The mass is rarely measured online, it is copied from a datasheet or changed by a payload, and a few-percent error biases the reading. Worse, the mass is only *conditionally identifiable*: in near-hover flight it is indistinguishable from a vertical external force, so an estimator that calibrates it unconditionally corrupts the very reading it supports. A virtual force sensor for aerial use therefore needs to know *when* its calibration is reliable and to be able to recalibrate itself when it is not. That is the problem we address.

Online parameter identification for aerial robots has been studied with maximum-likelihood frameworks [5] and with recursive regression on inertial signals that recover the mass and inertia under physical consistency constraints [6], [7]; these methods need excited flight to identify the inertial parameters. External-force estimation has a parallel line of work based on momentum observers [1], [2], which estimate the force on the airframe from the dynamics but are biased by errors in the same parameters above. Moving-horizon estimation has been applied to quadrotor system identification [4], building on the constrained-MHE theory [3]; the unmodeled aerodynamics that learned residual models capture are themselves an active area [10], [11]. On the control side, adaptive and incremental schemes reject disturbances without an explicit force estimate [9], and learning-based adaptive controllers fit a disturbance representation online to reject wind in agile flight [13]. These methods cancel the disturbance but do not reason about *when* the underlying parameters are identifiable.

Two themes run through this work. First, the mass is recoverable only under sufficient excitation, a persistency-of-excitation condition that classical optimal input design addresses by shaping the input to maximize the information about the unknown [8]. Freezing a parameter update when it is not persistently excited is itself classical in adaptive estimation, in the form of dead-zone and covariance-management schemes [9], and exciting the input to restore information is classical optimal input design [8]. Our contribution is not either element alone but the signal that drives both: unlike a dead-zone keyed to a generic excitation proxy, and unlike a momentum observer that estimates the force without quantifying its identifiability, we gate on the closed-form Schur complement, which (i) isolates exactly the mass–vertical-force coupling, (ii) is the marginal Fisher information on the mass (its inverse is the Cramér–Rao bound), and (iii), in its dimensionless form $\tilde{\sigma}$, doubles as the controller's trigger to restore observability. Second, the external force and the inertial parameters are coupled, so estimating them jointly can bias both. The near-hover mass–vertical-force degeneracy we use is a known structural fact in aerial estimation; existing methods, to our knowledge, do not quantify it online to decide *when* a parameter is reliable or to *act* so that it becomes observable. In near-hover flight the mass and the vertical external force enter the dynamics through the same channel and are not jointly observable, so an estimator that updates both distributes the error between them. This is the property we exploit.

This paper builds an observability-aware, self-calibrating virtual force sensor. The external force enters the accelerometer almost algebraically, so the sensor reads it accurately once the mass is calibrated; the mass, by contrast, is only conditionally identifiable and is confounded with a vertical force near hover. We make the calibration observability-aware on two levels. An extended Kalman filter (EKF) monitors an online identifiability measure and freezes the mass calibration when the parameter is unobservable, protecting the reading during hover where a naive filter drifts. When the mass must be recalibrated but the flight is too weakly exciting to reveal it, the model-predictive controller injects a minimal probe that restores identifiability and then stops, so the sensor never relies on a parameter it cannot see. Fig. 1 shows the resulting loop and the role of the identifiability signal $\tilde{\sigma}$.

The contributions of this work are the following. The bare reading of the external specific force from the accelerometer residual is a known momentum/IMU-residual technique [1], [2], [9]; our contribution is the observability-aware self-calibration layer that makes such a virtual force sensor reliable on a vehicle whose calibration parameter is only con-

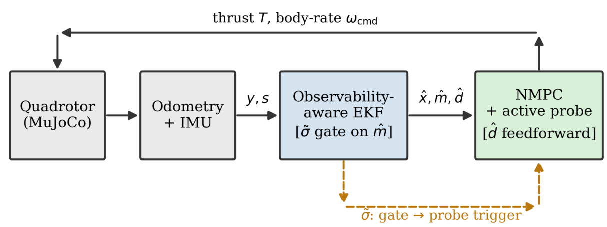

FIGURE 1. Architecture of the self-calibrating virtual force sensor: the identifiability σ gates the mass calibration and triggers the active probe; the NMPC feeds $\hat{d}$ forward.

ditionally identifiable.

- We give the observability analysis of the sensor's calibration: the mass and the vertical external force are not jointly observable in near-hover flight, and we derive an online identifiability measure, the Schur complement of the force block in the thrust–acceleration regressor Gramian, which is the marginal Fisher information on the mass (its inverse is the Cramér–Rao bound) and quantifies when the calibration is reliable.
- We make the sensor *self-calibrating*. The EKF gates the mass calibration by this measure, freezing it when unobservable so the reading is protected during hover; and when the mass must be recalibrated but the flight lacks sufficient excitation, the controller injects a probe to restore identifiability. The same $\tilde{\sigma}$ *shapes* the probe, since maximizing it selects the most informative excitation within the probe family, so a single observability scalar governs gating, triggering, and excitation design.

The resulting sensor is characterized and validated end to end in software-in-the-loop: it matches the simulator ground truth to about $0.4\,\mathrm{N}$ RMS per axis (per-axis correlation 0.93 under continuous wind), a moving-horizon estimator does not improve on it because the reading is algebraically observable, and feeding the sensed force to a model-predictive controller reduces the tracking error by 34 to $80\,\%$ under wind, in real time at $100\,\mathrm{Hz}$.

This is a methodology and feasibility study in software-in-the-loop. We exercise the sensor against injected accelerometer bias, scale-factor, and thrust-mismatch errors to probe the dominant real-sensor effects, but a full hardware demonstration, on a real IMU and thrust map, or on a manipulator whose joint-torque readings give the same observability structure, is the natural next step and is deliberately left to a dedicated follow-up.

The rest of the paper is organized as follows. Section II presents the translational model and the measurement equations. Section III analyzes observability and derives the hover ambiguity and the identifiability measure. Section IV describes the observability-aware EKF and its mass gate. Section V presents the disturbance-rejecting controller and the active excitation. Section VI reports the experimental setup and Section VII the results. Section VIII concludes.

II. SYSTEM MODEL

We model the quadrotor as a rigid body and estimate only its translational motion. The attitude is measured by the onboard estimator and enters the model as a known input, which removes the orientation states from the estimator and keeps it well-conditioned.

A. FRAMES AND NOTATION

Let $\mathcal{F}_W = \{O_W, \boldsymbol{x}_W, \boldsymbol{y}_W, \boldsymbol{z}_W\}$ be the inertial world frame and $\mathcal{F}_B = \{O_B, \boldsymbol{x}_B, \boldsymbol{y}_B, \boldsymbol{z}_B\}$ the body frame attached to the center of mass (CoM). The position of $\mathcal{F}_B$ with respect to $\mathcal{F}_W$ is $\boldsymbol{p} \in \mathbb{R}^3$ and its velocity is $\boldsymbol{v} \in \mathbb{R}^3$. The orientation of $\mathcal{F}_B$ with respect to $\mathcal{F}_W$ is the rotation matrix $\mathbf{R}(\boldsymbol{q}) \in SO(3)$, parameterized by the unit quaternion $\boldsymbol{q} = [q_0,\ \boldsymbol{q}_v^\top]^\top \in \mathbb{S}^3$, with $\mathbb{S}^3 \triangleq \{\boldsymbol{q} \in \mathbb{H} : \|\boldsymbol{q}\| = 1\}$ following the convention of [14]. The body rate $\boldsymbol{\omega} \in \mathbb{R}^3$ is the angular velocity of $\mathcal{F}_B$ with respect to $\mathcal{F}_W$ expressed in $\mathcal{F}_B$. We write $[\cdot]_\times$ for the skew-symmetric operator that maps a vector to the matrix form of the cross product. The thrust acts along $\boldsymbol{z}_B$, so its direction in $\mathcal{F}_W$ is $\boldsymbol{a} \triangleq \mathbf{R}(\boldsymbol{q})\,\boldsymbol{e}_3$ with $\boldsymbol{e}_3 = [0,0,1]^\top$.

B. RIGID-BODY DYNAMICS

The quadrotor is a six-degree-of-freedom rigid body actuated by a collective thrust T along the body z axis and by an inner-loop body-rate controller. Its state is the position $\boldsymbol{p}$, velocity $\boldsymbol{v}$, attitude $\boldsymbol{q}$, and body rate $\boldsymbol{\omega} \in \mathbb{R}^3$. The dynamics are

$$\dot{\boldsymbol{p}} = \boldsymbol{v}, \tag{1}$$

$$\dot{\boldsymbol{v}} = -g\,\boldsymbol{e}_3 + \frac{T}{m}\mathbf{R}(\boldsymbol{q})\,\boldsymbol{e}_3 + \boldsymbol{d} - c\,\|\boldsymbol{v}\|\,\boldsymbol{v}, \tag{2}$$

$$\dot{\boldsymbol{q}} = \tfrac{1}{2}\,\boldsymbol{\Omega}(\boldsymbol{\omega})\,\boldsymbol{q}, \tag{3}$$

$$\dot{\boldsymbol{\omega}} = \tfrac{1}{\tau_{rc}}\,(\boldsymbol{\omega}_{\text{cmd}} - \boldsymbol{\omega}), \tag{4}$$

where g is the gravitational acceleration, $m \in \mathbb{R}_{>0}$ is the mass, and c is a quadratic drag coefficient that lumps the dominant unmodeled aerodynamics [12]. The attitude kinematics (3) use the matrix $\boldsymbol{\Omega}(\boldsymbol{\omega}) = \left[\begin{smallmatrix} 0 & -\boldsymbol{\omega}^\top \\ \boldsymbol{\omega} & -[\boldsymbol{\omega}]_\times \end{smallmatrix}\right]$, following the convention of [14], [15]. The body rate follows the first-order response (4) to the commanded rate $\boldsymbol{\omega}_{\text{cmd}}$, with time constant τ_{rc} for the inner-loop rate controller, while the collective thrust T is commanded directly. The term $\boldsymbol{d} \in \mathbb{R}^3$, expressed in $\mathcal{F}_W$, is the *external specific force*, the net external force per unit mass acting at the center of mass,

$$\boldsymbol{d} \triangleq \frac{1}{m}\boldsymbol{f}_e, \tag{5}$$

where $\boldsymbol{f}_e \in \mathbb{R}^3$ is the resultant of the forces that the nominal model does not represent: aerodynamic effects beyond the drag term, wind, and any added payload. Equivalently, $\boldsymbol{d}$ is the residual between the true acceleration and the acceleration that gravity and thrust predict in (2); the experiments set the explicit drag coefficient $c = 0$, so $\boldsymbol{d}$ also lumps the aerodynamic drag. We treat $\boldsymbol{d}$ as quasi-static over the estimation window in near-hover flight, where the velocity-dependent drag is negligible and an added mass and a vertical force become indistinguishable (Section III); under fast flight $\boldsymbol{d}$ varies with velocity, which the random-walk process model tracks within its bandwidth. Thus $\boldsymbol{d}$ is a *lumped specific-force residual*: the external force (wind, payload) plus the lumped aerodynamic drag, the former dominating near hover and the latter growing with speed. The force estimate is validated against the simulator's external-force channel, and the residual drag is what the random-walk bandwidth limits at high speed (Section VII). The quantities estimated online are the mass m and the external force $\boldsymbol{d}$, which enter the translational dynamics (2); the rate-loop constant τ_{rc} is a nominal known constant of the inner loop, not estimated.

For estimation the attitude is treated as a known input: $\boldsymbol{q}$, and hence the thrust axis $\boldsymbol{a}$, are taken from the accurate onboard estimate, so the estimator integrates only the translational dynamics (1)–(2), with the thrust T as a known input. This avoids the quaternion states, with their re-normalization and hemisphere bookkeeping, and keeps the estimator well-conditioned over the window.

C. MEASUREMENT MODEL

The onboard sensors provide several signals at each step: the position and velocity from odometry, the attitude $\boldsymbol{q}$ and the body rate $\boldsymbol{\omega}$ from the inertial estimator and the gyroscope, and the specific force from the accelerometer. The translational estimator uses two of them and treats the attitude as an input. The first is the position–velocity measurement

$$\boldsymbol{y} = [\boldsymbol{p}^\top,\ \boldsymbol{v}^\top]^\top \in \mathbb{R}^6, \tag{6}$$

provided by the odometry, which is noisy in the simulator. The second is the accelerometer, which measures the specific force in $\mathcal{F}_B$, the non-gravitational acceleration sensed by the IMU [16]; rotated into $\mathcal{F}_W$ with the measured attitude it reads

$$\boldsymbol{s} = \mathbf{R}(\boldsymbol{q})\,\boldsymbol{a}_{\text{IMU}} = \frac{T}{m}\boldsymbol{a} + \boldsymbol{d} - c\,\|\boldsymbol{v}\|\,\boldsymbol{v}, \tag{7}$$

the non-gravitational part of (2). The attitude $\boldsymbol{q}$ enters as the known input through $\boldsymbol{a} = \mathbf{R}(\boldsymbol{q})\,\boldsymbol{e}_3$ rather than as a state to estimate, and the body rate $\boldsymbol{\omega}$ is not needed by the translational estimator. Because the thrust T and the attitude $\boldsymbol{q}$ are known, (7) is an algebraic measurement of $\boldsymbol{d}$: the accelerometer residual senses the external force directly, which turns the estimator into a virtual force sensor.

III. OBSERVABILITY AND EXCITATION

The accelerometer measurement (7) relates the unknowns to known signals, so we analyze when the mass and the external force can be recovered from it. The drag term is a known function of the measured velocity, so we move it to the left-hand side and define the compensated specific force $\tilde{\boldsymbol{s}} \triangleq \boldsymbol{s} + c\,\|\boldsymbol{v}\|\,\boldsymbol{v}$. With the inverse mass $\beta \triangleq 1/m$, (7) becomes

$$\tilde{\boldsymbol{s}} = T\,\boldsymbol{a}\,\beta + \boldsymbol{d} = \boldsymbol{\Phi}\,\boldsymbol{\varphi}, \qquad \boldsymbol{\Phi} = [\,T\boldsymbol{a}\ \ \mathbf{I}_3\,], \quad \boldsymbol{\varphi} = [\beta,\ \boldsymbol{d}^\top]^\top, \tag{8}$$

which is linear in the unknown $\boldsymbol{\varphi} \in \mathbb{R}^4$ with regressor $\boldsymbol{\Phi} \in \mathbb{R}^{3\times 4}$. The thrust T and the thrust axis $\boldsymbol{a}$ are known, so identifiability of $\boldsymbol{\varphi}$ is the identifiability of $(m, \boldsymbol{d})$.

The accelerometer is what makes (8) algebraic. Without it, the force $\boldsymbol{d}$ would enter only through the velocity derivative $\dot{\boldsymbol{v}}$ in (2) and would have to be recovered by differentiating the noisy odometry twice, which amplifies noise and adds a delay. Reading $\boldsymbol{d}$ from the specific force (7) removes that differentiation and yields an instantaneous algebraic measurement, which is the basis of the virtual force sensor and decouples the analysis of $\boldsymbol{d}$ from the integration of the trajectory.

The estimator augments the position and velocity with the mass and the force. The position and velocity are measured by the odometry (6) at every step, so the recoverability of the parameters reduces to the *identifiability* of $(m, \boldsymbol{d})$ from the accelerometer model (8) accumulated over the window, a persistency-of-excitation condition on the regressor rather than a full nonlinear observability test of the augmented state. This identifiability determines when the mass and the force can be separated, and is what we analyze and exploit.

A single sample gives three equations for four unknowns, so φ is recovered only by accumulating samples. Over a window of N samples the information is the Gramian $\mathbf{G} = \sum_{k=1}^{N} \boldsymbol{\Phi}_k^\top \boldsymbol{\Phi}_k$, which has the block form

$$\mathbf{G} = \begin{bmatrix} \sum_k T_k^2 & \sum_k T_k \boldsymbol{a}_k^\top \\ \sum_k T_k \boldsymbol{a}_k & N\,\mathbf{I}_3 \end{bmatrix}, \tag{9}$$

where $\|\boldsymbol{a}_k\| = 1$ was used. The force block $N\mathbf{I}_3$ is always nonsingular, so given the mass $\boldsymbol{d}$ is recovered whenever the window is nonempty; the joint degeneracy that remains is confined to the mass–vertical-force pair at hover (Proposition 1). The mass enters only through β, which is identifiable if and only if the Schur complement of the force block is positive,

$$\sigma \triangleq \sum_k T_k^2 - \tfrac{1}{N} \left\| \sum_k T_k \boldsymbol{a}_k \right\|^2 > 0. \tag{10}$$

The scalar σ is the dispersion of the vectors $T_k \boldsymbol{a}_k$ about their mean; it is zero only when they are identical across the window, which is the case at hover. Its dimension is thrust squared (newtons squared, not to be confused with the window length N), so its absolute scale grows with the thrust level of the maneuver. To obtain a dimensionless, maneuver-comparable measure we normalize it by the thrust energy in the window,

$$\tilde{\sigma} \triangleq \frac{\sigma}{\sum_k T_k^2} \in [0, 1], \tag{11}$$

the relative dispersion of the thrust direction: $\tilde{\sigma} = 0$ at hover and $\tilde{\sigma} \to 1$ when the thrust vector is maximally spread. Dividing out the absolute thrust level removes the dominant source of the threshold's maneuver dependence, so a single $\tilde{\sigma}_{\min}$ classifies the regime across maneuvers far better than the raw σ, as Section VII quantifies.

Proposition 1. *In stationary level hover, where $\boldsymbol{a}_k = \boldsymbol{e}_3$ and $T_k = \bar{T}$ are constant over the window, the inverse mass β is not identifiable ($\sigma = 0$). Equivalently, only the combination $\bar{T}\beta + d_z$ is observable, while β and the vertical force d_z are not separately observable.*

Proof. With $\boldsymbol{a}_k = \boldsymbol{e}_3$ and $T_k = \bar{T}$ we have $\sum_k T_k^2 = N\bar{T}^2$ and $\sum_k T_k \boldsymbol{a}_k = N\bar{T}\boldsymbol{e}_3$, so $\sigma = N\bar{T}^2 - \frac{1}{N}N^2\bar{T}^2 = 0$, and β is not identifiable by (10). The structure is explicit in (8): the vertical row reads $\tilde{s}_z = \bar{T}\beta + d_z$, a single equation in the two unknowns β and d_z, whereas the horizontal rows read $\tilde{s}_{x,y} = d_{x,y}$ since $\boldsymbol{a} = \boldsymbol{e}_3$. Hence d_x and d_y are observable while β and d_z collapse onto their sum. □

Remark 1. *The ambiguity is confined to the vertical axis and to the pair (m, d_z): a mass error and a vertical force produce the same accelerometer residual at hover. The joint ambiguity is confined to the mean thrust direction, which at and near hover is vertical, so the horizontal components d_x and d_y remain observable there, which is why the external force is recovered in the directions where wind dominates.*

Away from hover the thrust axis tilts and its magnitude varies, so the vectors $T_k \boldsymbol{a}_k$ disperse and $\sigma > 0$; the window is then persistently exciting for the mass. Since the Cramér–Rao lower bound on the mass estimate scales with σ^{-1} (derived in (12) below), a more aggressive trajectory admits a better-conditioned estimate, while the force, conditioned by the block $N\mathbf{I}_3$ given the mass, is read with the same accuracy in every regime with an accurate held mass. The mass is thus observable under motion and degenerate only at hover, which Section VII confirms by identifying it accurately across the range of flight speeds.

The scalar σ is therefore an online measure of how well the mass can be identified, read directly from the known thrust and attitude over the window with no extra sensing. It also has an estimation-theoretic meaning. With accelerometer noise of variance r_s, the joint Fisher information of $(\beta, \boldsymbol{d})$ over the window is $\mathbf{G}/r_s$ with $\mathbf{G}$ the Gramian (9). The $(1, 1)$ block of its inverse is the inverse Schur complement, $[\mathbf{G}^{-1}]_{11} = 1/\sigma$, so the *marginal* Fisher information of the inverse mass β, after profiling out the jointly estimated force $\boldsymbol{d}$, is σ/r_s, and the Cramér–Rao bound on any unbiased estimate is

$$\operatorname{var}(\hat{\beta}) \;\geq\; r_s\,[\mathbf{G}^{-1}]_{11} \;=\; r_s\,\sigma^{-1}, \tag{12}$$

which diverges as $\sigma \to 0$ at hover and tightens as the maneuver disperses the thrust vector. The bound (12) is the batch least-squares CRB under zero-mean accelerometer noise of isotropic covariance $r_s\mathbf{I}_3$ and a quasi-static $\boldsymbol{d}$ over the window; it is the limit the recursive filter approaches under excitation, used here as a relative identifiability measure rather than an absolute uncertainty. Equation (12) gives the *raw* scalar σ an estimation-theoretic meaning: σ/r_s is the marginal Fisher information on the mass, so σ is the inverse Cramér–Rao bound (up to the noise variance). The bound is stated on the inverse mass β; since $\operatorname{var}(\hat{m}) \approx m^4 \operatorname{var}(\hat{\beta})$ for small errors, the bound on the mass itself scales with the same σ^{-1}. For *gating* we use its dimensionless normalization $\tilde{\sigma}$ (11), which divides out the thrust level so that a single threshold $\tilde{\sigma}_{\min}$ holds across maneuvers; σ carries the information content and $\tilde{\sigma}$ carries the decision. We do not invert (12) to set $\tilde{\sigma}_{\min}$ analytically; it is tuned with margin between the hover

TABLE 1. Observability regimes (Proposition 1). The "degraded" d_z entry is the joint case; with an accurate held mass it is recovered (Section VII).

Quantity	Hover	Maneuver
mass m	unobservable ($\sigma \approx 0$)	observable (σ large)
d_x, d_y	observable	observable
d_z	degraded (m–d_z)	observable

and maneuver values of $\tilde{\sigma}$ (Section VII), and (12) reports the uncertainty that any admissible threshold tolerates. The estimator uses $\tilde{\sigma}$ to decide when to trust the mass, and the controller of Section V uses it to decide when to excite.

Proposition 1 has a direct operational reading, summarized in Table 1. The external force is recoverable in every regime given an accurate held mass, and is recovered by any recursive estimator that uses the accelerometer residual; the mass is recoverable only when the flight excites it (σ large), and is then identified jointly with the force. This motivates the two elements of our method: an estimator that updates the mass only when σ is large (Section IV), and a controller that creates the excitation when the mass must be identified but the flight lacks the excitation to reveal it (Section V).

IV. OBSERVABILITY-AWARE ESTIMATION

We estimate the translational state and the parameters of Section II with an extended Kalman filter (EKF) whose state augments the position and velocity with the mass and the external force,

$$\boldsymbol{x} = [\boldsymbol{p}^\top,\, \boldsymbol{v}^\top,\, m,\, \boldsymbol{d}^\top\,]^\top \in \mathbb{R}^{10}. \tag{13}$$

The known inputs at each step are the thrust T and the thrust axis $\boldsymbol{a}$. The position and velocity follow (1)–(2), while the mass and the external force have no dynamics of their own and evolve as random walks with diagonal process covariance $\mathbf{Q} = \mathrm{diag}(q_p\mathbf{I}_3, q_v\mathbf{I}_3, q_m, q_d\mathbf{I}_3)$ (Table 2). Treating the attitude as a known input keeps the filter free of quaternion states and their log-map singularity; the body rate is not needed by the translational estimator.

A. PREDICTION AND UPDATE

Each step propagates the dynamics, $\hat{\boldsymbol{x}}^- = \boldsymbol{f}(\hat{\boldsymbol{x}})$ and $\mathbf{P}^- = \mathbf{F}\mathbf{P}\mathbf{F}^\top + \mathbf{Q}$ with $\mathbf{F} = \partial \boldsymbol{f}/\partial \boldsymbol{x}$, and applies two measurement updates. The odometry (6) is a linear update through $\mathbf{H} = [\mathbf{I}_6\ \mathbf{0}]$ with covariance $\mathbf{R}$. The accelerometer (7) is a nonlinear update with model $\boldsymbol{g}(\boldsymbol{x}) = (T/m)\boldsymbol{a} + \boldsymbol{d} - c\,\|\boldsymbol{v}\|\,\boldsymbol{v}$, Jacobian $\mathbf{H}_s = \partial \boldsymbol{g}/\partial \boldsymbol{x}$, and covariance $\mathbf{R}_s$; this residual is the term that drives $\boldsymbol{d}$ and makes the filter a virtual force sensor. Both updates use the standard EKF gain $\mathbf{K} = \mathbf{P}^-\mathbf{H}^\top(\mathbf{H}\mathbf{P}^-\mathbf{H}^\top + \mathbf{R})^{-1}$, and the mass and force are clipped to physical bounds after each step.

B. THE MASS GATE

A plain EKF updates every state at every step, including the mass when it is unobservable. Under the hover ambiguity of Proposition 1 this attributes a vertical force to a mass change and corrupts the estimate. The observability-aware filter instead conditions the mass update on the identifiability $\tilde{\sigma}$ of (11), computed online over a sliding window from the known thrust and attitude. When the mass is not persistently excited, the corresponding row of the Kalman gain is set to zero and its process noise is suppressed,

$$\mathbf{K}_m \leftarrow \mathbf{0}, \quad q_m \leftarrow 0 \qquad \text{while} \quad \tilde{\sigma} < \tilde{\sigma}_{\min}. \tag{14}$$

With $q_m = 0$ and the mass having no dynamics of its own ($F_{mm} = 1$), the prediction leaves the mass variance unchanged, $P_m^- = P_m$, so while the gate is closed the mass neither drifts nor inflates: it is held exactly at its last well-identified value with its uncertainty. The realized hold is therefore as good as the gate's closure: in still hover the gate stays shut and the mass is held exactly, whereas a windy hover intermittently re-excites $\tilde{\sigma}$ and re-opens the gate, leaving the residual movement reported in Section VII. Suppressing the process noise is what makes this hold; were the variance left to grow, a transient spike in $\tilde{\sigma}$ would re-open the update with a large gain and let a single sample move the mass. To reject such spikes the gate re-opens the update only after $\tilde{\sigma}$ has stayed above $\tilde{\sigma}_{\min}$ for a dwell time t_d (Table 2), not on an isolated sample; this separates the sustained excitation of a maneuver or probe from a momentary gust. The force update is never gated, since, given the held mass, $\boldsymbol{d}$ is observable throughout (Section III).

C. CHOICE OF ESTIMATOR: EKF VERSUS MOVING HORIZON

The accelerometer makes $\boldsymbol{d}$ an algebraic measurement (8), so the force needs no temporal smoothing to be recovered, and the mass, once excited, is identified by the recursive update alone. A moving-horizon estimator over the same state and measurements was implemented for comparison and did not improve the estimates (Section VII): the window adds cost without a benefit when the informative measurement is instantaneous and the recursive filter already carries the full measurement history. We therefore retain the EKF and place the contribution in the observability logic rather than in the estimator class.

V. CLOSED-LOOP USE: CONTROL AND ACTIVE CALIBRATION

The virtual sensor is used in closed loop by a nonlinear model-predictive controller (NMPC) that serves two roles: it rejects the sensed disturbance by feedforward (the application), and it injects the active probe that recalibrates the mass when needed (the self-calibration of Section V-B). We first describe the controller, then the probe.

The estimated force closes the loop through this NMPC, which tracks a reference trajectory while rejecting the disturbance by feedforward. The controller keeps the full pose, so its state is the position, velocity, attitude, and body rate, $\boldsymbol{x}_c = [\boldsymbol{p}^\top, \boldsymbol{v}^\top, \boldsymbol{q}^\top, \boldsymbol{\omega}^\top]^\top \in \mathbb{R}^{13}$, and its input is the collective thrust and the commanded body rate, $\boldsymbol{u} = [T,\, \boldsymbol{\omega}_{\text{cmd}}^\top]^\top \in \mathbb{R}^4$.

The prediction model is the rigid-body dynamics (1)–(4) with the estimated mass and the estimated disturbance injected,

$$\dot{\boldsymbol{v}} = -g\,\boldsymbol{e}_3 + \frac{T}{\hat{m}}\,\mathbf{R}(\boldsymbol{q})\,\boldsymbol{e}_3 + \hat{\boldsymbol{d}}, \tag{15}$$

so that $\hat{\boldsymbol{d}}$ enters as an additive specific-force feedforward that cancels the external force; the rate loop keeps the form (4) with the nominal time constant τ_{rc}. The prediction model carries no explicit drag term; because the estimator runs with $c = 0$, the aerodynamic drag is lumped into $\hat{\boldsymbol{d}}$ and is therefore compensated through the same feedforward. The disturbance estimate is smoothed by an exponential moving average before injection (the mass estimate is smoothed likewise, with the constants of Table 2). In the closed-loop experiments the online update is the force alone: the mass in the prediction model is held at its identified value (here the known $1.05\,\mathrm{kg}$), so it is the feedforward $\hat{\boldsymbol{d}}$ that closes the loop, while the mass calibration protects the force reading rather than retuning the controller. Because the gate acts only on the mass row of the filter, the force estimate of the gated and the plain filter coincide, and the injected $\hat{\boldsymbol{d}}$ is that common reading.

A. COST AND CONSTRAINTS

Over a horizon of N_c nodes the controller minimizes

$$J = \sum_{k=0}^{N_c-1} \Big(\big\|\boldsymbol{p}_k - \boldsymbol{p}_k^{\mathrm{ref}}\big\|^2_{\mathbf{Q}_p} + \big\|\boldsymbol{v}_k - \boldsymbol{v}_k^{\mathrm{ref}}\big\|^2_{\mathbf{Q}_v} + \|\boldsymbol{e}_q(\boldsymbol{q}_k)\|^2_{\mathbf{Q}_a} + \big\|\boldsymbol{u}_k - \boldsymbol{u}_k^0\big\|^2_{\mathbf{R}_u} \Big) + \|\boldsymbol{e}_N\|^2_{\mathbf{Q}_N}, \tag{16}$$

where $\boldsymbol{p}^{\mathrm{ref}}$ and $\boldsymbol{v}^{\mathrm{ref}}$ are the reference position and velocity, $\boldsymbol{e}_q = \mathrm{Log}(\boldsymbol{q}^{-1} \otimes \boldsymbol{q}^{\mathrm{ref}}) \in \mathbb{R}^3$ is the attitude error through the quaternion logarithm, with the standard hemisphere correction, and the input is regularized around the hover equilibrium $\boldsymbol{u}^0 = [\hat{m}g,\, \mathbf{0}^\top]^\top$ so that a wrong nominal mass does not bias the regularization. The velocity reference $\boldsymbol{v}^{\mathrm{ref}}$ is a feedforward term that removes the tracking lag at high speed; $\mathbf{R}_u$ weights the input deviation. The terminal error $\boldsymbol{e}_N$ stacks the position, velocity, and attitude errors at stage N_c, weighted by $\mathbf{Q}_N = \mathrm{diag}(\mathbf{Q}_p, \mathbf{Q}_v, \mathbf{Q}_a)$. The inputs are bounded by $0 \le T \le T_{\max}$ and $\|\boldsymbol{\omega}_{\mathrm{cmd}}\|_\infty \le \omega_{\max}$, with $T_{\max} = 49\,\mathrm{N}$ ($\approx 4.8mg$) and $\omega_{\max} = 20\,\mathrm{rad\,s^{-1}}$.

The problem is solved with sequential quadratic programming under the real-time iteration scheme, HPIPM, and a Gauss–Newton Hessian within acados [17], sharing the framework with the estimator. It uses $N_c = 31$ nodes over a $1.5\,\mathrm{s}$ horizon at $100\,\mathrm{Hz}$, with a per-step solve time below $1\,\mathrm{ms}$.

B. ACTIVE EXCITATION

The gate of Section IV-B protects a mass that has already been identified, but it cannot create the information the flight does not provide: in sustained hover the mass stays unobservable and the estimate is frozen at the prior it started from. When an up-to-date mass is required and the identifiability is low, the controller closes this gap by actively exciting the unobservable direction. The same identifiability that gates the estimate now *shapes* the excitation: by (12) the Schur complement is the Fisher information on the mass, so the probe that maximizes it (approximately, its dimensionless form $\tilde{\sigma}$, since the probe perturbs the thrust energy only to second order about $T \approx mg$) per unit deviation minimizes the mass uncertainty. We do not solve the continuous optimal-input-design problem [8]; instead we maximize $\tilde{\sigma}$ over a small two-parameter probe family by an empirical sweep, the most informative probe within that family. Proposition 1 identifies the two levers: a vertical bob of amplitude δ_z that modulates the thrust magnitude T, and a lateral motion of amplitude δ_ℓ that tilts the thrust axis $\boldsymbol{a}$. Both disperse the regressor $T\boldsymbol{a}$, so the probe is

$$\boldsymbol{p}^{\mathrm{ref}} \leftarrow \boldsymbol{p}^{\mathrm{ref}} + \big[\delta_\ell(\cos\omega_\ell t - 1),\ \delta_\ell \sin\omega_\ell t,\ \delta_z \sin\omega_z t\big]^\top, \quad \tilde{\sigma} < \tilde{\sigma}_{\min}, \tag{17}$$

with $\omega_\ell = 2\pi f_\ell$ and $\omega_z = 2\pi f_z$ (Table 2), and we choose the (δ_z, δ_ℓ) mix that maximizes $\tilde{\sigma}$ subject to a fixed amplitude budget $\delta_z^2 + \delta_\ell^2 \le \rho^2$, identified empirically in Section VII. The probe is triggered only while the mass is both needed and unobservable; once the estimate has converged it is withdrawn and the gate again protects the mass. This is the active counterpart of the gate.

C. REMARKS ON CLOSED-LOOP BEHAVIOR

The model error reaching the controller stays bounded. The uncompensated force is the estimation residual $\boldsymbol{d} - \hat{\boldsymbol{d}}$, small because $\boldsymbol{d}$ is observable throughout, and the mass enters only as the multiplicative thrust scaling $\hat{m}/m$, whose error is bounded by (12) where the mass is observable and frozen at its identification value where it is not. A formal stability analysis of the coupled estimator–controller loop is left to future work.

VI. EXPERIMENTAL SETUP

We evaluate the virtual sensor and its self-calibration in software-in-the-loop; this section describes the simulation environment, the flights and disturbance, the baselines, and the metrics.

A. SIMULATION ENVIRONMENT

All experiments run in software-in-the-loop with the MuJoCo physics engine [18]. The quadrotor has a mass of $1.05\,\mathrm{kg}$, the sum of the body geometries in the model description, which we take as the ground truth for the mass estimate. The odometry is published with injected noise on position and velocity, so the estimator is exercised on noisy measurements, and the simulator exposes the true external force applied to the airframe on a dedicated channel, which serves as ground truth for the force estimate. The estimator, controller, and the scripts that reproduce every figure will be released on publication, together with a data dictionary that maps each logged column to its physical meaning and the source file for every figure and table. The parameters are listed in Table 2.

TABLE 2. Estimator and controller parameters.

Block	Parameter	Value
Plant / sim	mass, rate $1/\tau_{rc}$	$1.05\,\mathrm{kg}$, $18\,\mathrm{s}^{-1}$
Drag	c (estimator)	0 (lumped into $\hat{\boldsymbol{d}}$)
Odometry noise	injected on p, v	MuJoCo, released config
Wind	profile, amplitude	step / sine, $\sim 2\,\mathrm{N}$
EKF meas. var.	r_p, r_v, r_s	$0.02,\ 0.1,\ 0.5$
EKF process	q_p, q_v, q_m, q_d	$10^{-6}, 10^{-3}, 10^{-5}, 10^{-2}$
Mass gate	window, dwell t_d, $\tilde{\sigma}_{\min}$	$1.0\,\mathrm{s}$, $0.25\,\mathrm{s}$, 0.009
Smoothing	α_m, α_d	$0.995,\ 0.92$
Active probe	budget ρ, f_ℓ, f_z, optimal	$0.40\,\mathrm{m}$, $0.5\,\mathrm{Hz}$, $1.0\,\mathrm{Hz}$, lateral
NMPC	N_c, horizon	31, $1.5\,\mathrm{s}$
NMPC weights	$\mathbf{Q}_p, \mathbf{Q}_a, \mathbf{Q}_v$	$180,\ 20,\ 30$
	$\mathbf{R}_u$	$\mathrm{diag}(0.3, 5, 5, 5)$
NMPC limits	$T_{\max}$, $\omega_{\max}$	$49\,\mathrm{N}$, $20\,\mathrm{rad\,s}^{-1}$

B. CONTROLLER, TRAJECTORIES, AND DISTURBANCE

The vehicle is flown by the NMPC of Section V at $100\,\mathrm{Hz}$, the same rate as the estimator. Two references are used: a stationary hover and a Lissajous figure-eight flown at peak speeds from 2.1 to $10.2\,\mathrm{m\,s}^{-1}$, which span the excitation levels of the observability analysis. The external force is a *deterministic* per-axis step profile of about $2\,\mathrm{N}$, commanded by the controller on the simulator's external-force channel; the simulator applies it and re-publishes the *applied* wrench as ground truth. Because the profile is defined in code, the disturbance is identical across runs and its onset is controlled (it is armed only once the vehicle has settled, after the verified reset), which makes every experiment reproducible. The force estimate, by contrast, is read from the accelerometer residual, independent of the commanded value, so the comparison against ground truth is not circular. Sustained steps keep the disturbance within the bandwidth of the momentum-based reading, examined in Section VII. For the rejection study the same disturbance is applied to the runs with and without the feedforward.

C. BASELINE

The baseline is the *plain EKF*, the same filter of Section IV but without the mass gate, run on the identical measurement stream. This ablation isolates the effect of the observability logic, since the two filters differ only in the gate. As a check on the estimator class, we also ran a moving-horizon estimator over the same state and measurements; it did not improve the estimates (Section IV-C) and is not carried as a separate baseline.

D. METRICS

The virtual force sensor is assessed by the Pearson correlation between the estimated force and the ground truth, per axis. Mass identification is assessed by the error relative to the ground truth and its spread across repetitions. Disturbance rejection is assessed by the position tracking error, reported as the root-mean-square error with and without the feedforward under the same wind, and its significance is tested with a one-sided Mann–Whitney U test at each speed. Each condition uses five repetitions with and five without the feedforward, so the pooled estimation statistics are $N = 10$ per speed; the hover feedforward case has four (one run failed the reset check and was dropped), making it $N = 4$ for the hover rejection point and $N = 9$ for the pooled hover estimation. We report mean $\pm$ std and the per-step solve time. The gate and active-excitation figures (Figs. 7, 8) are each reported over five repetitions with independent noise and wind realizations, shown as mean $\pm$ std; the disturbance-free mass-identification figure (Fig. 5) is a single representative run.

VII. RESULTS

We present the results in the order of the contributions: the external force, the mass identification and its hover limit, the gate, the active excitation, the disturbance rejection, robustness, and a summary comparison against both baselines.

A. THE VIRTUAL FORCE SENSOR

The accelerometer makes the force algebraic (8), so the EKF recovers it without any temporal smoothing. We report three complementary measures so that no single one is misleading: the absolute error, the per-axis Pearson correlation against the step ground truth (Fig. 3), and the correlation under a continuously varying disturbance (Section VII-G). The absolute error is $0.4\,\mathrm{N}$ root-mean-square per axis on a $\sim 2\,\mathrm{N}$ disturbance, about $20\,\%$ of the applied force; the calibration curve below shows this residual is dominated by the bandwidth of the random-walk force model, not by a scale error. The step correlation is 0.91 to 0.98 across speeds (inflated by the step transitions, hence reported alongside the absolute error); and under a sinusoidal wind the correlation is 0.93 (Section VII-G), the most demanding case. The high-speed rows are also conservative in a specific sense: the ground-truth channel contains only the applied wind, while the aerodynamic drag is additionally lumped into $\hat{\boldsymbol{d}}$ (c=0 in the estimator), so the correlations at speed mix the two; near hover, where the drag vanishes, the reading is validated cleanly. A moving-horizon estimator matches the EKF on these and does not improve them, confirming that the force needs no special estimator beyond the accelerometer residual. Fig. 2 shows the estimate tracking the disturbance in the time domain on all three axes.

The reading itself is the translational analogue of a classical momentum/residual observer [1], [2], and we do not claim to improve it: any such observer computed with the same mass would return the same force. The comparison that matters is therefore not against another force estimator but against the same estimator without the calibration layer, which is the ablation of the following subsections; Fig. 6 quantifies exactly the mass bias that an uncalibrated, or unconditionally calibrated, observer inherits.

We characterize the sensor with a calibration curve, the standard test for a force sensor: the estimated force ($m\hat{\boldsymbol{d}}$) against the applied force, per axis, over a sinusoidal wind that sweeps the range (Fig. 4). The fit is linear and correctly scaled, with slopes 0.98, 0.97, 0.96 for the three axes, sub-

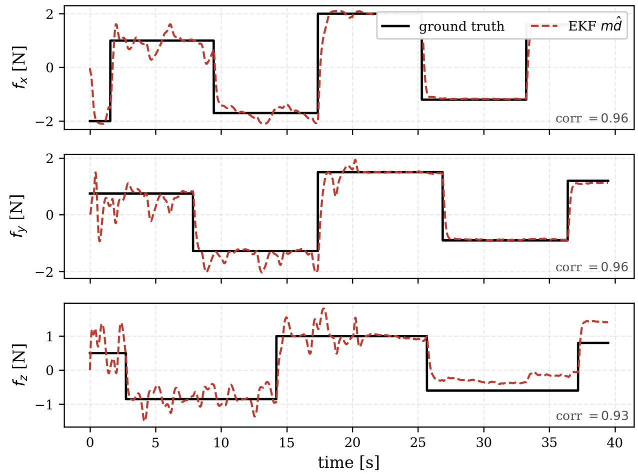

FIGURE 2. Virtual force sensor in the time domain (representative run): $m\hat{d}$ tracks the applied wind steps on all three axes.

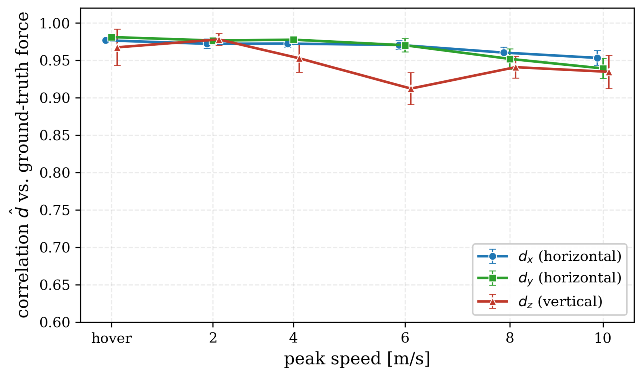

FIGURE 3. Per-axis force correlation against excitation (*N*=10; 9 in hover; step ground truth).

$0.2\,\mathrm{N}$ intercepts, and R^2 of 0.90, 0.89, 0.87; the unity slope confirms the reading is correctly scaled by the mass and the small residual is the bandwidth limit of the random-walk force model. The sensor is bandwidth-limited rather than range-limited: an amplitude sweep of step disturbances shows the slope holding above 0.8 up to about $2\,\mathrm{N}$ and then falling (to 0.4 at $4\,\mathrm{N}$) as large, fast steps exceed the random-walk bandwidth, so the usable full scale is set by how quickly the disturbance varies, a limitation a higher-order force model would relax.

B. MASS IDENTIFICATION AND ITS HOVER LIMIT

Given excitation and no confounding disturbance, the mass is identified accurately. Fig. 5 shows the EKF converging from a deliberately wrong prior ($0.60\,\mathrm{kg}$, $43\,\%$ low) to within $0.6\,\%$ of the ground truth on a disturbance-free trajectory, with the estimated force correctly near zero. This is the identification result: the mass is a physical invariant and is recovered to the true value once the trajectory excites the thrust direction.

Hover is where identification is hard, and it exposes the limit of the ungated filter. Fig. 6 shows the plain EKF mass

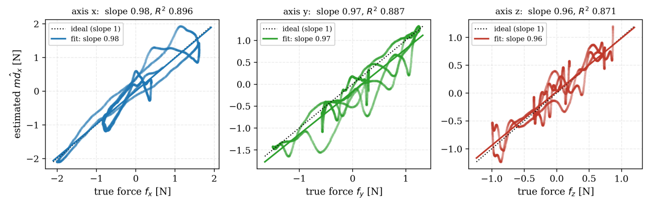

FIGURE 4. Calibration curve: estimated force $m\hat{d}$ versus applied force, per axis, under a sinusoidal wind.

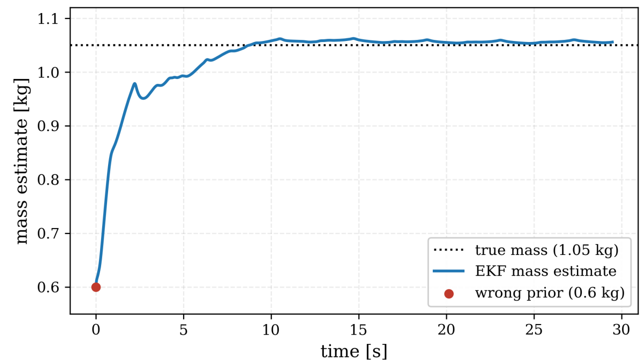

FIGURE 5. Mass identification under excitation: from a $0.60\,\mathrm{kg}$ prior to within $0.6\,\%$ (disturbance-free trajectory).

against speed under wind *without* the gate: in hover the estimate is biased by $4.8\,\%$ and varies by $\pm 35\,\mathrm{g}$ across the $N = 9$ runs, because the vertical force is misread as a mass change (Proposition 1); under excitation the bias falls below $1\,\%$ and the spread to $\pm 2\,\mathrm{g}$, in agreement with the σ^{-1} conditioning of Section III. This hover bias, which appears even when the mass is seeded from the correct value, is exactly what the gate prevents, as the next experiment shows.

C. THE MASS GATE PROTECTS THE ESTIMATE

Fig. 7 flies an aggressive trajectory followed by a hover, both under wind, over five runs with independent noise and wind. During the trajectory both filters identify the mass. In the hover that follows, the plain EKF is blind to the lost observability and attributes the vertical wind to a mass change; its drift is more pronounced here than in the steady-hover battery of Fig. 6 ($4.8\,\%$), because the maneuver leaves the filter with an inflated mass covariance, so the wind moves the estimate by $7\,\%$ over the transition (7.0 ± 0.6, $N{=}5$). The observability-aware filter detects $\tilde{\sigma}$ falling below $\tilde{\sigma}_{\min}$ and gates the mass update, cutting the drift to $3\,\%$ (3.3 ± 0.8); a residual movement remains because the windy hover still excites $\tilde{\sigma}$ intermittently, but the gate more than halves the error and is consistently better than the plain EKF across all runs. The gate protects an estimate that hover can no longer fully support.

This advantage comes specifically from the Schur complement, not merely from gating on excitation. We ran the same filter with the gate keyed instead to a generic persistency-of-excitation proxy, the total regressor energy $\sum_k \|\boldsymbol{\Phi}_k\|^2$,

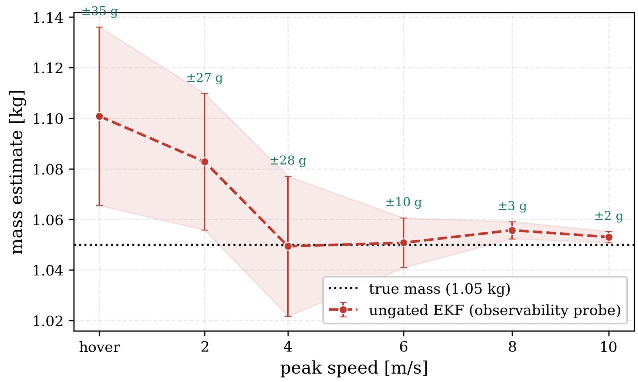

FIGURE 6. Mass observability against excitation, *ungated* EKF (*N*=10 per speed, *N*=9 in hover; mean ± std).

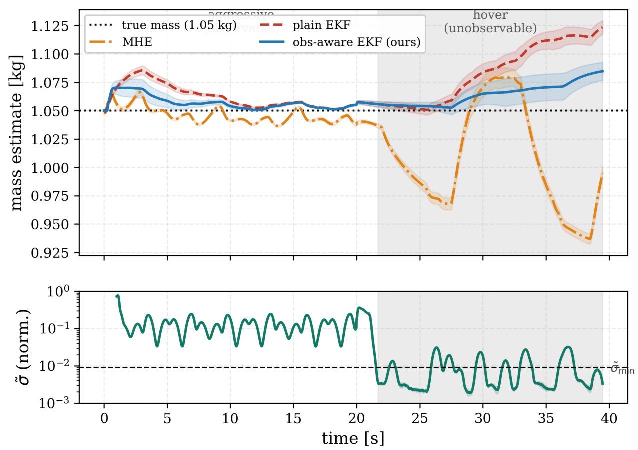

FIGURE 7. Mass over a windy maneuver-to-hover transition (*N*=5, mean ± std); hover shaded, σ and threshold at the bottom.

a conventional dead-zone. Because hover is high in thrust energy though zero in dispersion, this proxy never closes and the result coincides with the ungated plain EKF (7.0 %): a generic excitation measure cannot tell hover from a maneuver. Only $\tilde{\sigma}$, which measures the dispersion that isolates the mass–vertical-force coupling, detects the unobservability and protects the calibration.

D. ACTIVE EXCITATION RECOVERS AN UNOBSERVABLE MASS

When the mass is unknown and the vehicle must hover, no passive estimator can recover it. Fig. 8 starts from a deliberately wrong prior ($0.70\,\mathrm{kg}$) in hover. The passive filter stays frozen at the prior, since hover provides no information (the passive estimate stays at $-33.3\,\%$ across all $N{=}5$ runs); the active controller injects the probe of (17), $\tilde{\sigma}$ rises above the threshold, the gate opens, and the mass converges from 0.70 toward the ground truth, after which the probe is withdrawn and the estimate is again protected. Using the most informative probe (Section VII-E), the mass error falls from $33\,\%$ to about $3\,\%$ (2.5 ± 1.6, $N{=}5$); the recovery depends on the probe shape, which the next experiment optimizes. By

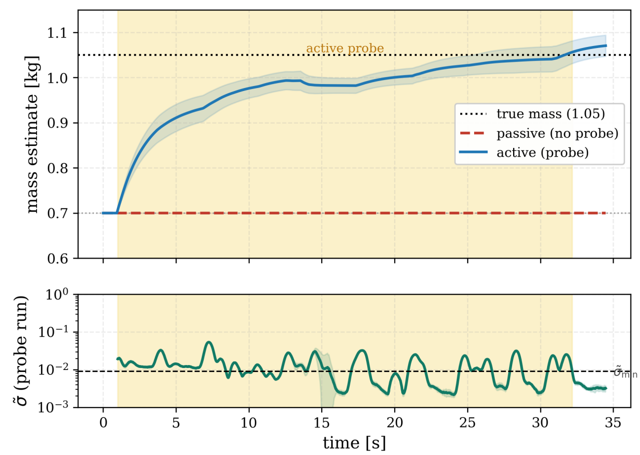

FIGURE 8. Mass recovery from a wrong hover prior ($0.70\,\mathrm{kg}$; *N*=5, mean ± std): the probe (shaded) lifts σ, recovers the mass, then withdraws.

contrast, under a disturbance-free trajectory the same filter identifies the mass to within $0.6\,\%$ (Fig. 5); the residual here is the price of identifying in hover from a wrong prior.

E. SHAPING THE PROBE BY IDENTIFIABILITY

The recovery in Fig. 8 depends on the probe shape, and the identifiability tells us which shape is best: since the Schur complement is the Fisher information on the mass (12), the probe that raises it most per unit deviation tightens the bound. We sweep the vertical/lateral mix of (17) at a fixed amplitude budget, from a pure lateral tilt to a pure vertical bob, and measure the achieved $\tilde{\sigma}$ and the resulting mass recovery (Fig. 9, $N{=}5$ per shape). The lateral tilt raises $\tilde{\sigma}$ well above every other mix and is the only shape whose recovery is both accurate and tightly distributed, consistent with (12): it is a variance bound, so the highest-$\tilde{\sigma}$ probe admits the lowest-variance recovery. The most informative mix (a lateral tilt) recovers the mass to $2.5\,\%$, against $8\,\%$ for the original mixed shape at the same amplitude budget; the difference is significant against every other shape (one-sided Mann–Whitney, $p < 0.01$). This shape is optimal only within the two-parameter family we sweep, not the unconstrained optimal input.

F. DISTURBANCE REJECTION

Feeding the estimated force to the controller reduces the tracking error under wind, and the reduction scales with how much the wind dominates the dynamics (Fig. 11, Table 4). Up to $6\,\mathrm{m\,s^{-1}}$ the wind dominates and the feedforward cuts the error by 77 to $80\,\%$ ($78\,\%$ in hover); as the speed grows the wind becomes a small fraction of the aerodynamic forces and the reduction falls to $34\,\%$ at $10\,\mathrm{m\,s^{-1}}$. The improvement is statistically significant at every speed: a one-sided Mann–Whitney U test of the feedforward against the baseline rejects the null at each operating point ($p < 0.01$, $n = 5$ per condition; $p = 0.008$ for the $n = 4$ hover case), with a large pooled effect size (Cohen's $d = 5.7$). The estimator and

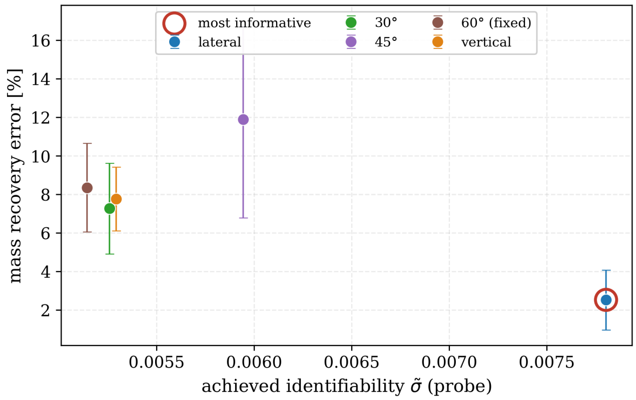

FIGURE 9. Probe shaping: mass recovery error against the vertical/lateral mix at a fixed amplitude budget; the σ-maximizing mix is circled.

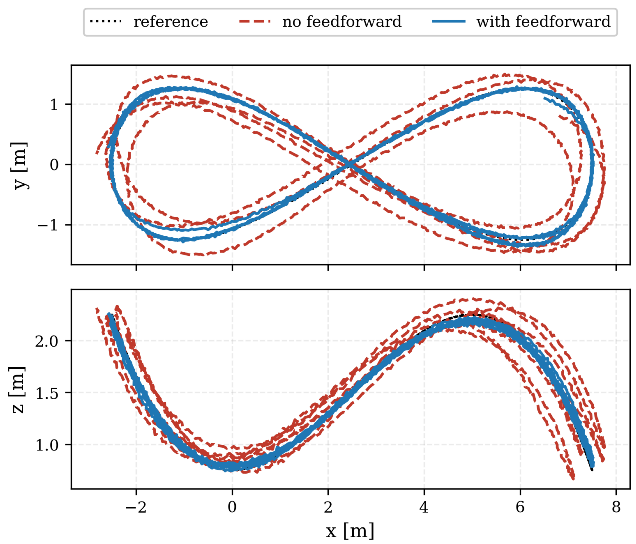

FIGURE 10. Trajectory under wind: with the feedforward (blue) versus without (red); reference dotted.

controller each run at 100 Hz with a per-step time below 1 ms. Fig. 10 shows the spatial effect on the figure-eight: with the feedforward the vehicle stays on the reference, while without it the wind pushes it off the path.

G. ROBUSTNESS

The virtual force sensor is not specific to the step disturbance used above. Under a sinusoidal wind, a continuously varying profile, the estimate tracks the ground truth with a per-axis correlation near 0.93 (Fig. 12), slightly below the step value because the moving disturbance has more bandwidth. All results so far use the simulator's injected odometry noise, so the estimator is already exercised on noisy measurements rather than a clean ground-truth state. The gate threshold is stable across maneuvers once the identifiability is normalized. The raw σ separates the regimes but its absolute scale follows the thrust level, so it required a maneuver-specific threshold (a span of more than an order of magnitude across

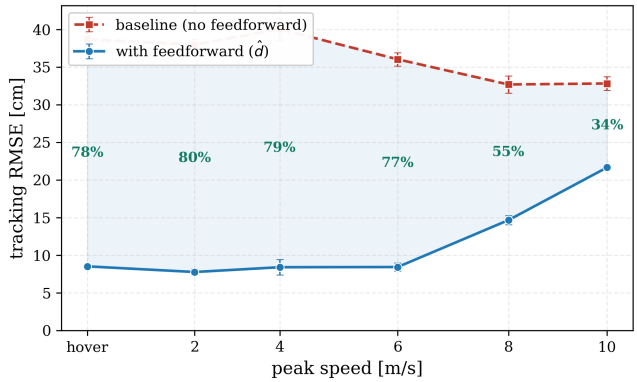

FIGURE 11. Disturbance rejection against speed (N=5 per mode; hover N=4; mean $\pm$ std).

TABLE 3. EKF estimation against trajectory speed (N=10; 9 in hover): mass estimate and per-axis force correlation with the ground truth. The force is read accurately throughout; the (ungated) mass is biased and noisy in hover and accurate under excitation.

Peak speed [m s^{-1}]	$\hat{m}$ [kg]	corr d_x/d_y	corr d_z
hover	1.101 ± 0.035	0.98/0.98	0.97
2.1	1.083 ± 0.027	0.97/0.98	0.98
3.8	1.049 ± 0.028	0.97/0.98	0.95
6.1	1.051 ± 0.010	0.97/0.97	0.91
8.2	1.056 ± 0.003	0.96/0.95	0.94
10.2	1.053 ± 0.002	0.95/0.94	0.93

our experiments). The dimensionless $\tilde{\sigma}$ of (11) removes that dependence: settled hover sits at $\tilde{\sigma} \approx 0.005$ (windowed, peak ≈ 0.007) while an aggressive trajectory reaches $\tilde{\sigma} \approx 0.34$, nearly two orders of magnitude apart. A single $\tilde{\sigma}_{\min} = 0.009$ classifies hover, maneuver, and the active probe alike: the probe lifts the instantaneous $\tilde{\sigma}$ above the threshold often enough to re-open the gate (Fig. 9 reports the run median, which sits just below it, with the lateral tilt the highest of the family), while hover keeps the gate closed. The remaining sensitivity is to transient gusts during a windy hover transition, which can momentarily spike $\tilde{\sigma}$; the dwell time t_d, which requires the threshold to be exceeded for a sustained interval, rejects these so the gate holds. The threshold is thus dimensionless and shared across all regimes, with the dwell, not the absolute signal scale, handling the residual gust sensitivity.

H. ROBUSTNESS TO SENSOR ERRORS

A virtual force sensor lives or dies by the error sources a real IMU and thrust map introduce, which the ideal simulator does not exercise. We inject them into the estimator inputs and sweep their magnitude (Fig. 13): an accelerometer bias up to 0.4 m s^{-2}, a scale-factor error up to 10 %, and a thrust-coefficient mismatch up to 10 %. The sensor degrades gracefully and proportionally, with no breakdown. The force error rises from 0.42 N to at most 0.58 N (under the largest bias), and the mass calibration from below 1 % to at most 11.5 % (under a 10 % thrust mismatch, which couples almost one-

TABLE 4. Disturbance rejection (N = 5 per mode; hover N = 4): tracking RMSE with and without the force feedforward under the same wind, against peak speed.

Peak [$\mathrm{m\,s^{-1}}$]	No ff [cm]	ff [cm]	Reduction
hover	38.7	8.5	78%
2.1	38.0	7.8	80%
3.8	40.0	8.4	79%
6.1	36.0	8.4	77%
8.2	32.7	14.6	55%
10.2	32.8	21.6	34%

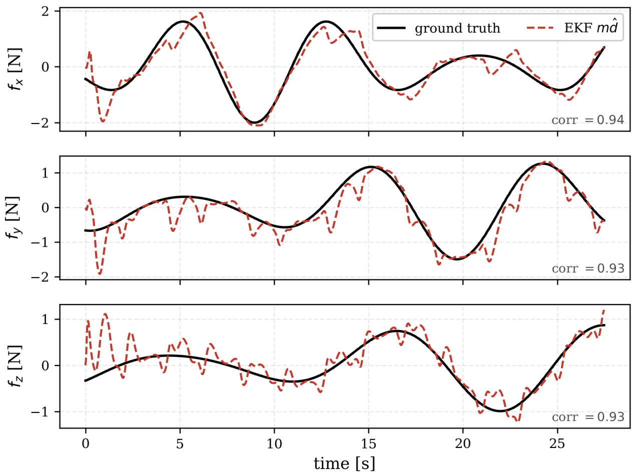

FIGURE 12. Virtual force sensor under a sinusoidal wind (representative run); per-axis correlation near 0.93.

to-one into the mass). The bias loads the force but barely the mass, while the scale and thrust errors load the mass through the T/m term, as expected; the near-linear degradation indicates the calibration error is recoverable by a one-time bench calibration of the bias and scale.

I. ESTIMATOR COMPARISON

Table 5 consolidates the comparison on the identical data. The two baselines fail in different ways, both traceable to the hover ambiguity of Proposition 1. The plain EKF reads the force well, including the vertical axis, but lets the hover mass drift as the vertical wind is misread as a mass change. The moving-horizon estimator is worse: its free mass both corrupts the vertical force, which it misreads as mass, and drifts in hover (Fig. 7), and the window adds cost without recovering either. Our observability-aware EKF keeps both: the EKF structure holds the vertical force, and the gate keeps the mass closest to the truth in the windy hover ($3.3\,\%$, against $7.0\,\%$ for the plain EKF and $5.3\,\%$ for the MHE). With active excitation it is also the only one that recovers the mass from a wrong prior in hover. All three run in real time at $100\,\mathrm{Hz}$.

VIII. CONCLUSION

We built a self-calibrating virtual force sensor for quadrotors. The external specific force is read directly from the accelerometer; its only calibration parameter, the mass, is observable only under excitation and is confounded with a vertical force in hover. We made the calibration observability-aware on two levels: an extended Kalman filter that gates the mass calibration by the online identifiability $\tilde{\sigma}$, freezing it when it cannot be seen, and a model-predictive controller that injects a minimal probe to restore identifiability when the mass must be recalibrated but the flight lacks the excitation. In software-in-the-loop the sensor reads the external force to about $0.4\,\mathrm{N}$ RMS per axis (per-axis correlation 0.93 under continuous wind), the gate holds the mass across a windy hover transition to $3\,\%$ where a plain EKF drifts to $7\,\%$, the active probe recovers the mass from a wrong prior that hover alone cannot identify, and the force feedforward reduces the tracking error by 34 to $80\,\%$ under wind, scaling with how much the wind dominates the dynamics.

A matched comparison showed that a moving-horizon estimator does not improve on the EKF for this problem, because the informative measurement is instantaneous; the contribution therefore lies in the observability logic, not in the estimator class. The dead-zone and the excitation are each classical; what ties them together is the closed-form Schur complement, the Fisher information on the mass computed from known signals (its inverse is the Cramér–Rao bound), whose dimensionless form $\tilde{\sigma}$ drives the gate and the probe from one online scalar, including the choice of the most informative excitation. The main limitation is that the study is in simulation; hardware validation with a motion-capture system and a calibrated disturbance is the immediate next step, as is a continuous (rather than family-restricted) optimal-input design. Because the filter reads any external specific force, the approach extends toward contact and tether forces, where an observability-aware wrench estimate would support aerial physical interaction without a dedicated force sensor.

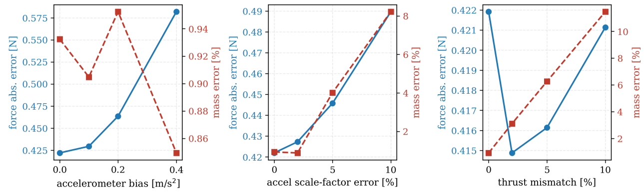

FIGURE 13. Robustness to injected accelerometer bias, scale-factor error, and thrust mismatch: force error (blue, left axis) and mass error (red, right axis).

TABLE 5. Estimator comparison on identical data. Force entries are pooled over the battery; the excited-mass entries pool the well-excited runs (8.2–10.2 $\mathrm{m\,s^{-1}}$, N=10 each speed); hover and recovery entries are from Figs. 7 and 8.

	MHE	plain EKF (baseline)	obs-aware EKF (ours)
horizontal force corr	0.96	0.97	0.97
vertical force corr	0.66	0.95	0.95
mass error, excited	+0.4%	+0.4%	+0.4%
mass error, windy hover	5.3%	7.0%	**3.3%**
recover mass, wrong prior	no	no	yes

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